Pool-Frenkel conduction (Sahoo et al., 2012)

↳ $\varepsilon = 6.48$ (Hu et al., 2017)

→ $\beta = \sqrt{\frac{q^3}{\pi \varepsilon \varepsilon_0}}$ → $\mu = 200 \frac{cm^2}{Vs}$ (Hu et al., 2017)

→ N_c = effective density of states in CB

$N_c = 4 \times 10^4$ (Hu et al., 2017)

But N_t from Hu et al. (2017) is in a.u. so

$N_c = 4^* 10^4$ a.u. → constant of proportionality

From Park (2020): $N_c = 2.94 E18\ cm^{-3}$

→ $J = q N_c \mu \left(\frac{V}{L}\right) \exp\left(\frac{\beta \sqrt{V/L} - q\phi_t}{kT}\right)$

→ $q\phi_t = V_0 - q\chi$

$q\phi_t = 2.45$ eV (Ambrosio et al., 2019)

$= 2.5$ eV (Srivastava et al., 2016)

SCLC (Murgatroyd, 1969)

Proportion of charge wich is free:

$$\frac{\rho_F}{\rho_F + \rho_t} = \theta_0 = \quad (A)$$

ρ_t: Trapped charge

→ E_A: Below the bottom of CB (Hu 2017)

$E_A = 0.5\,eV$

Poisson's equation:

→ $\frac{dE}{dx} = \frac{\rho_t + \rho}{\varepsilon \varepsilon_0}$ (B)

→ $J = \mu \rho_t E$ (C)

Frankel effect:

→ $\frac{\rho_F}{\rho_F + \rho_t} = \frac{N_c}{N_t} \exp\left[-\frac{E_A}{KT} + \frac{1}{kT} B\sqrt{E}\right]$ (D)

→ $B = \sqrt{\frac{q^3}{\pi \varepsilon \varepsilon_0}}$

Frankel effect is important if:

$$\frac{1}{kT} B\sqrt{\frac{V}{L}} > \frac{E_A}{KT}$$

From (A), (B), (C) y (D)

$$J = \mu \theta_0 \varepsilon \varepsilon_0 E \exp(B E^{1/2}) \frac{dE}{dx}$$

$$J = \frac{9}{8} \mu \varepsilon \varepsilon_0 \frac{V^2}{L^3} \theta_0 \exp\left[\frac{0.89}{kT}\sqrt{\frac{e^3 V}{\pi \varepsilon \varepsilon_0 L}}\right]$$

$$= \frac{9}{8} \mu \varepsilon \varepsilon_0 \frac{V^2}{L^3}\left[\frac{N_c}{N_t} \exp\left(-\frac{E_A}{KT}\right)\right] \exp\left[\frac{0.89}{kT}\sqrt{\frac{e^3 V}{\pi \varepsilon \varepsilon_0 L}}\right]$$

For ZnO:

$$\rightarrow \frac{1}{kT}\beta\sqrt{E} = \frac{\sqrt{E}}{0.02585\,eV}\sqrt{\frac{(1e)^3}{\pi(6.48)(553\times10^3\,e/Vcm)}}$$

$$= 19.53 * 10^3 \frac{1}{eV}\sqrt{\frac{e^3\,Vcm}{e}}\sqrt{E}$$

$$= 19.53\times10^3\sqrt{\frac{cm}{Volts}}\sqrt{\frac{V}{L}}$$

$$\rightarrow \frac{E_A}{kT} = \frac{0.5\,eV}{0.02585\,eV} = 19.34$$

Thus:

$$\frac{1}{kT}\beta\sqrt{\frac{V}{L}} > \frac{E_A}{kT}$$

$$19.53\times10^3\sqrt{\frac{cm}{Volts}}\sqrt{\frac{V}{L}} > 19.34$$

$$\sqrt{\frac{V}{L}} > \frac{19.34}{11.53\times10^3}\sqrt{\frac{Volts}{cm}}$$

$$V > L\left(\frac{19.34}{11.53*10^3}\right)^2\frac{Volts}{cm}$$

$$V > (10\times10^{-7}\,cm)\left(\frac{19.34}{11.53*10^3}\right)^2\frac{Volts}{cm}$$

$$V > 2.81\times10^{-12}\,Volts$$